# Cambridge (CIE) IGCSE Computer Science
# Paper 1: Computer Systems (Set B)

**Tuesday 6 May 2025**

**Morning (Time: 1 hour 45 minutes)**

Total marks

**/ 75**

**Instructions**

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.
- Calculators must not be used in this paper.

**Scan here to mark your mock exam**
or visit the mock exams landing page for this course

1 (a) All data needs to be converted to binary data so that it can be processed by a computer.

**Explain** why a computer can only process binary data.

..............................................................................................................................................................

..............................................................................................................................................................

**(2 marks)**

(b) The denary values 64, 101 and 242 are converted to 8-bit binary values.

**Give the 8-bit binary value** for each denary value.

64 ...............................

101 ...............................

242 ...............................

..............................................................................................................................................................

..............................................................................................................................................................

..............................................................................................................................................................

**(3 marks)**

(c) The hexadecimal values 42 and CE are converted to binary.

**Give the binary value** for each hexadecimal value.

42 .................

CE .................

..............................................................................................................................................................

..............................................................................................................................................................

..............................................................................................................................................................

..............................................................................................................................................................

**(4 marks)**

**2 (a)** The Unicode **character set** is used to represent text that is typed into a computer.

**Describe** what is meant by a character set.

..........................................................................................................................................

..........................................................................................................................................

**(2 marks)**

**(b)** **Discuss** the trade-offs between **image quality** and **file size** when choosing a bitmap image's resolution and colour depth.

Provide **examples** to illustrate your answer.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

**(8 marks)**

**3 (a)** A student has a sound file that is too large to be stored on their external secondary storage device. The student compresses the sound file to make the file size smaller.

The compression method used **reduces the sample rate** and the **sample resolution** of the sound file.

**Identify** which **type** of compression has been used to compress the sound file.

..............................................................................................................................

**(1 mark)**

**(b)** The following data is stored as a **text file**:

| |
|---|
| red, green, yellow, green, purple, blue, red, purple, blue, yellow, grey, black, pink, red, |

**Explain** how lossless compression would compress this file.

..............................................................................................................................

..............................................................................................................................

..............................................................................................................................

..............................................................................................................................

..............................................................................................................................

**(5 marks)**

**4** **Complete the statements** about different types of software.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

- application
- assembly language
- bootloader
- central processing unit (CPU)
- firmware
- hardware
- operating
- output
- system
- user

.............................................................................. software provides the services that the computer requires; an example is utility software.

.............................................................................. software is run on the operating system.

The .............................................................................. system is run on the firmware, which is run on the....................................................

........................................................................................................................................................................................

........................................................................................................................................................................................

........................................................................................................................................................................................

........................................................................................................................................................................................

**(4 marks)**

5 (a) **Describe** the difference between an **opcode** and an **operand.**

**(2 marks)**

(b) A computer has a Von Neumann architecture

**Describe** the **purpose of the control unit** (CU) within this computer.

**(2 marks)**

(c) A computer has a Von Neumann architecture

The computer has a **single core CPU**.

The computer is upgraded to a **dual core CPU**.

**Explain how** the upgrade can **affect the performance** of the computer

**(2 marks)**

**6 (a)** A web server has an internet protocol (IP) address.

**Identify** the network component that uses the IP address to send data only to its correct destination.

..................................................................................................................................

**(1 mark)**

**(b)** Many devices have a Media Access Control (MAC) address.

**Give three** features of a MAC address.

..................................................................................................................................

..................................................................................................................................

..................................................................................................................................

**(3 marks)**

**7 (a)** A company has a website that is stored on a web server.

The company is concerned about a distributed denial of service (**DDoS**) attack

**Suggest one** security device that can be used to help prevent a DDoS attack.

..............................................................................................................................

**(1 mark)**

**(b)** A student is concerned about the threats to their computer when using the internet

The student wants to use some security solutions to help protect the computer from the threats.

**Identify** a security solution that could be used to protect the computer from a **computer virus**, **hacking** and **spyware**.

Each security solution **must** be different.

..............................................................................................................................

..............................................................................................................................

..............................................................................................................................

**(3 marks)**

**8 (a)** Robots are used in a factory to build cars.

One characteristic of a robot is its mechanical structure.

**State two** other characteristics of a robot.

..................................................................................................................................................

..................................................................................................................................................

**(2 marks)**

**(b)** **Suggest two** advantages of using robots, instead of humans, to build cars in the factory.

..................................................................................................................................................

..................................................................................................................................................

**(2 marks)**

**9 (a)** Three Internet terms are browser, Internet Protocol (IP) address and Uniform Resource Locator (URL).

**Five** statements are given about the Internet terms

**Tick (✓)** to show which statements apply to each Internet term. Some statements may apply to **more than one** Internet term.

| **Statement** | **Browser (✓)** | **IP address (✓)** | **URL (✓)** |
|---|---|---|---|
| it contains the domain name | | | |
| it is a type of software | | | |
| it converts Hypertext Markup Language (HTML) to display web pages | | | |
| it is a type of address | | | |
| it stores cookies | | | |

**(5 marks)**

**(b)** Joelle is a student who uses the Internet

The table contains five terms or definitions that relate to the Internet

**Complete the table** by writing each missing term or definition.

| Term | Definition |
|---|---|
| browser | |
| | this is the company that provides a user with a connection to the Internet |
| | this is a protocol that is used to send data for web pages across the Internet |
| Uniform Resource Locator (URL) | |
| cookie | |

**(5 marks)**

10 (a) Sensors and a microprocessor are used to maintain the correct conditions in a fish tank.

**Describe how** the sensors and the microprocessor are used to maintain the correct conditions in the fish tank.

**(4 marks)**

(b) Jamelia has a greenhouse that she uses to grow fruit and vegetables. She needs to make sure the temperature in the greenhouse stays between 25°C and 30°C (inclusive).

A system that has a temperature sensor and a microprocessor is used to maintain the temperature in the greenhouse. The system will:

- open a window and turn a heater off if it gets too hot
- close a window and turn a heater on if it gets too cold.

**Describe how** the system uses the temperature sensor and the microprocessor to maintain the temperature in the greenhouse.

**(8 marks)**

**11 (a)** An expert system is an example of artificial intelligence. One component of an expert system is a knowledge base.

**Identify three** other components that are present in an expert system.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

**(3 marks)**

**(b)** An expert system is an example of artificial intelligence. One component of an expert system is a knowledge base.

**Explain why** an expert system needs a knowledge base.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

**(3 marks)**